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Combustion Dynamics Diagnostics and Mitigation on a Prototype Gas Turbine Combustor

Bassam Mohammad, Keith McManus and Jeffrey Ruszczyk

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Technical Report Abstract Page

Title	Combustion Dynamics Diagnostics and Mitigation on a Prototype Gas Turbine Combustor		
Author(s)	Bassam Mohammad Keith McManus Jeffrey Ruszczyk	**Phone**	1 518-387-5472 1 518-387-6597 1 518-387-6507
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In the current research a ceramic liner is used while the previous published data was collected with a quartz liner. Typically, quartz is avoided due to reductions in effective flame temperature by radiation losses. Experiments showed that NO emissions was not affected by the combustor liner type. This agreement between the quartz and ceramic liners data indicates limited effect from the radiation heat losses on NOx emissions.

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COMBUSTION DYNAMICS DIAGNOSTICS AND MITIGATION ON A PROTOTYPE GAS TURBINE COMBUSTOR

Bassam S. Mohammad, Keith McManus, Jeffrey Ruszczyk
General Electric Global Research Center
Niskayuna, NY 12309, USA

Ahmed M Elkady, Mark A Mueller
General Electric Aviation
Evendale, OH 45215, USA

ABSTRACT

Combustion dynamics have detrimental effects on hardware durability as well as combustor performance and emissions. This paper presents a detailed study on the impact of combustion dynamics on NO_x and CO emissions generated from a prototype gas turbine combustor operating at a pressure of 180 psia with a pre-heat temperature of 720 F (E-class machine operating conditions). Two unstable modes are discussed. The first is an intermittent mode that emerges at flame temperatures near 2900°F, resulting in high NOx and CO emissions. With increasing fuel flow, NOx and CO emissions continue to increase until the flame temperature reaches approximately 3250°F, at which point the second acoustic mode begins to dominate. Operation in this second 750 Hz mode results in significantly reduced NOx and CO emissions. At pressures higher than 180 psia, the intermittent mode intensifies, leading to flashback at flame temperatures above 2850°F. In order to mitigate the intermittent mode, a second configuration of the combustor included an exit area restriction. The exit area restriction eliminated the intermittent mode, resulting in stable operation and low emissions over a temperature range of 2700-3200°F. A comparison of the NOx emissions, as function of flame temperature, with previous published data for perfectly premixed indicates that, while the low amplitude 750 Hz oscillations have little effect, the intermittent mode significantly increases emissions.

In the current research a ceramic liner is used while the previous published data was collected with a quartz liner. Typically, quartz is avoided due to reductions in effective flame temperature by radiation losses. Experiments showed that NO emissions was not affected by the combustor liner type. This agreement between the quartz and ceramic liners data indicates limited effect from the radiation heat losses on NOx emissions.

1 INTRODUCTION

In recent decades, gas turbine combustor development has been heavily influenced by increasingly stringent emissions requirements. The development of the Dry Low Emissions (DLE) and Dry Low NOx (DLN) technologies has coincided with a need to mitigate the thermo-acoustic combustion instabilities associated with lean premixed combustion. Combustion instabilities can result in high amplitude pressure oscillations due to coupling between heat release fluctuations and system acoustics that are primarily dependent on system geometry and boundary conditions. Fundamental understanding of these phenomena as well as development of effective mitigation strategies is of prime importance to both gas turbine manufacturers as well as end users since combustion dynamics can have detrimental effects on the hot-section component durability and engine operability and emissions.

Well-controlled, premixed combustion experiments are often utilized to evaluate emissions entitlement and flammability limits [1-5]. Perfectly premixed combustion experiments, in which the fuel and air are mixed upstream of the combustor inlet, avoid the complexity and uncertainty in mixedness that is present in industrial gas turbine combustion systems. Hence, it allows the isolation of the flame chemistry effects and provides the opportunity to validate detailed chemical kinetics mechanisms. Several authors have reported the results of perfectly premixed flames at typical gas turbine

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pressures and temperatures. Elkady et. al. studied the production of NO_x and CO under perfectly premixed conditions in a high-pressure quartz combustor [6]. Their work extended the experiments of Leonard and Correa [1] and included the effect of residence time. However, in premixed combustion, the propensity for combustion acoustics is high, and could have a dominant effect on emissions. Combustion dynamics is an area of active research [7-11]; a recent review discusses in details different instability modes as well as active and passive control strategies [12].

This paper presents a study of the effect of combustion dynamics on emissions and operability using a prototype gas turbine combustor. Two liners were tested, one with an open end with a constant cross-sectional area and a second having an area restriction at the exit.. The configuration with a constant area cross-section generated high amplitude intermittent pressure oscillations which significantly affected emissions and operability. The configuration with an 44% area reduction reduced the dynamics significantly and extended the combustor operability. The emissions data from both configurations is presented and the data is analyzed to provide insight on the effect of the dynamics mode on emissions and operability.

The experiments also shed light on the effect of heat losses on NOx production. This is done by comparing the results from the ceramic liners to the results published recently from a quartz liner.

2 NOMENCLATURE

P_3	=	Combustor inlet pressure
T_3	=	Combustor air/Fuel mixture inlet temperature
TC	=	Thermocouple
P	=	Pressure
$\Delta P/P$	=	Percentage pressure drop across Combustor
NO_x	=	Nitric Oxides
CO	=	Carbon Monoxide
O_2	=	Oxygen
CO_2	=	Carbon dioxide
H_2	=	Hydrogen
ppmvd	=	Particle per million volume dry
T_{flame}	=	Flame temperature based on O2 emissions
ms	=	millisecond
ppm/ms	=	Particle per million per millisecond
ft/s	=	Velocity in feet's/second

3 EXPERIMENTAL SETUP AND TEST CONDITIONS:

3.1 Experimental facility

The test setup is very similar to that of Elkady et. al [6]. The ASME stamped pressure vessel has a diameter of 16" with pressure and temperature ratings of 600 psia and 1100°F, respectively. Details of the pressure vessel with air, fuel, water and exhaust connections are shown in Figure 1. Air is supplied through two independent feeds for combustion and cooling which are metered separately. High-pressure, preheated combustion air is connected to a T-connection where the fuel is injected. Prior to entering the combustor, the fuel and air mixture passes through a tube bundle, as shown in Figure 2, having a length greater than 200 tube diameters to achieve condition close to perfect mixedness. A number of thermocouples are installed on the tube surface to monitor for auto-ignition and flashback. These T/C's, along with combustor inlet temperature (T3) T/C, are connected to an interlock system that shuts down the fuel supply if auto ignition or flashback occurs (in the tube bundle or at the nozzle inlet). The cooling air is directed to the outer side of the combustor to balance the pressure across the liner and to cool the metallic support sleeve surrounding the liner. The exhaust gases emanating from the liner are mixed with the spent cooling air in the exhaust plenum where cooling water is injected to reduce the exhaust temperatures upstream of a pressure control valve. The cooled exhaust gases are directed through the exhaust piping and released to the atmosphere.

The fuel used in this study is a typical pipeline natural gas having a methane content of approximately 96%. Preheated 750°F is utilized to provide both combustion and liner cooling air. The total preheated air mass flow rate ranges between 1-2.4 lbm/sec depending on the operating conditions. Flame temperatures are varied from lean blow-out (LBO) up to 3300 F. Combustion air is metered using a venturi with an accuracy of less than 1%. Fuel is metered using a Coriolis type mass flow meter with an accuracy of 0.5%. Both meters were calibrated to ensure high measurement accuracy.

3.2 Ceramic Combustor

The ceramic combustor assembly is illustrated in Figure 3. The assembly comprises an inlet manifold with a conical entrance, burner, dome flange, ceramic liner, liner sleeve, cooling air sleeve and a mounting plate to assemble the combustor to the high pressure vessel. The air fuel mixing tube is connected to the conical entrance of the inlet manifold to straighten the reactant stream prior to a perforated plate that stabilizes the flame in the combustor. Measurements of combustor inlet pressure (P3) and inlet temperature (T3) are made at the inlet cone.

Details of the perforated flame stabilizer are shown in Figure 4. The stabilizer also serves as a flame arrestor, preventing flashback of the flame into the inlet manifold. The flame arrestor tubes have a diameter of 0.06 inch while the flame stabilizer tubes have a diameter of 0.125" arranged in a manner similar to the one used by Leonard et al. [1] and Elkady et al. [6]. The perforated plate is made of Inconel 718 and has 36 holes arranged as shown in Figure 4. The tube spacing is ¼" with the center hole designed to fit a ¼ inch igniter (hydrogen torch).

The ceramic liner is installed in an Inconel 718 tube by shrink fitting the tube to the ceramic liner. The ceramic liner has a length of 9 inches and an internal diameter of 1.9 in. The liner sleeve is welded to the mounting flange to prevent leakage of cooling air into the combustor section. Leakages upstream of the flame holder are prevented by welding the inlet manifold to the flame holder. .

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The burner assembly is mounted on the mounting flange with five bolts. A metallic seal is used to prevent cooling air from entering the combustor.

3.3 Emissions Sampling

Combustion products are sampled using a water-cooled gas sampling probe placed on an axial traverse. The sample is then transferred to the sample-conditioning unit via heated line passing through a heated filter. A stream of the sample is then cooled, dried, and sent to a gas sampling system for CO_2, O_2, UHC, and CO measurement. The other stream remains heated before entering the NO_x analyzer to eliminate loss of NO_2 The gas sampling unit is calibrated using standard calibration gases. The analyzers are spanned prior to tests and then checked during and after the experiments to ensure no drift. A recent gauge repeatability and reproducibility analysis (gauge R&R) was performed on the NO_x analyzer confirming a measurement accuracy of +/- 1 ppm of the full range. In this report, NO_x emissions are corrected and reported to a 15% O_2 level by volume.

3.4 Test conditions and data acquisition

Experiments are conducted at P_3 of 180 and 260 psia. The combustor residence time is roughly 20 ms. The combustor pressure drop ($\Delta P/P3$) varied from 0.5 to 1.1% depending on operating conditions. The stabilizing jet velocity (calculated based on preheated air flow) varied in the range of 44-50 ft/s depending on the test conditions.

Real-time data is acquired at a frequency of 1Hz. Reported data point represent an average of 10 consecutive data samples. Maximum NO_x standard deviation is roughly ±0.8ppm (1.9% mean value). The reported average P_3 and T_3 has a typical standard deviation of ±1.2 psi and ±4°F respectively. Flame temperatures are calculated using fuel-air ratios based on metered air and fuel flows or the O2 and/or CO2 percentages in the sampled combustor products. The latter technique is valid when complete combustion occurs; i.e., at flame temperature above the LBO limits. In this study, the flame temperature based on O_2 and CO_2 typically match to within 10-15 degrees. In addition, the flame temperature based on metered air and fuel matches with the temperature based on O2 emissions within 1%. In this paper, reported flame temperatures are calculated based on O_2 concentrations. The uncertainty in flame temperatures is estimated to be ±40°F.

4 RESULTS AND DISCUSSIONS

4.1 Effect of Combustion Dynamics on Emissions and Operability

Figures 5 and 6 present time-averaged NOx and CO behavior as function of flame temperature. Three regimes of combustor operability are observed. The first regime corresponds to a stable combustion mode at flame temperatures up to 2940°F. The second regime is characterized by a highly unstable mode at flame temperatures between 2940-3200°F. The third regime is a stable combustion mode characterized by low amplitude 750Hz oscillation at flame temperatures higher than 3250°F. In the following discussion, the data is discussed in a chronological order as fuel flow was increased to raise the flame temperature from 2700°F up to 3300°F.

The combustor is ignited at 180 psia, preheat temperature of 720°F and a flame temperature of about 2700°F. The ignition process was very smooth and the NOx emissions were low as expected at such low flame temperatures. As more fuel is injected, the NOx increased and CO decreased until a temperature of about 2940°F was reached (Figures 5 and 6). An example of dynamic pressure amplitudes in this quiet regime is shown in Figure 7. It is apparent that the combustion process is taking place at a high degree of stability as demonstrated by the low amplitude of pressure oscillations. A water cooled camera installed in the pressure vessel provided images of the flame stability. The image captured in the quiet mode showed consistent illumination indicating a steady flame which is in agreement with the acoustic signal.

In the second regime, an intermittent thermo-acoustic instability mode is encountered as shown in Figure 8. This mode results in an unsteady flame behavior leading to a significant increase in NOx and CO emissions as shown in Figures 5 and 6. NOx emissions reached levels 4x those measured in the absence of the dynamic instability. At the same time, CO emissions increased by 30x.. As the temperature exceeds 2900°F, the unstable mode is manifest via unsteady illumination due to the flame oscillation. Adding more fuel increases the dynamic pressure amplitude of the intermittent mode from 3 psi peak-to-peak (p-p) to more than 5 psi p-p as shown in Figure 8. Most probably the unsteady flame motion creates hot spots generating NOx while simultaneously causing intermittent flame extinction causing high CO.

As the flame temperature exceeds 3200°F, the intermittent mode diminishes and a constant low-amplitude 750 Hz oscillation prevails. Once the transition occurs, a sharp decrease in NOx is observed and CO levels return to equilibrium levels. A time trace of the dynamic pressures during transition from the intermittent mode to the constant low amplitude mode is shown in Figure 9.

When the experiment was repeated at higher pressure, it was observed that at pressures above 300 psia , flashback occurred at flame temperatures lower than the transition to stable operation. This change in behavior is primarily due to the effect of the combustor operating pressure on the acoustic impedance. The results show that the intermittency of combustion oscillations changes flame structure, resulting in increased levels of NOx and CO. It is also apparent that higher operating pressures intensify the amplitude of oscillations.

4.2 Intermittent Mode Mitigation

In an attempt to suppress the intermittent mode and extend the window of operation, an area restriction at the combustor exit was added as shown in Figure 3. Repeating the experiments with area restriction eliminated the intermittent

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mode, resulting in well-behaved NOx and CO data trends as seen in Figures 5 and 6, respectively. Compared to unsteady operation, NOx emissions are decreased by approximately 80%. It appears that the exit area restriction reduced the communication between the combustor and the exhaust plenum. This restricted communication significantly altered the acoustic impedance and extended the window of operation throughout 2700-3200 F flame temperatures. It is noteworthy that with the exit area restriction the combustor operability was also extended to pressures higher than 400 psia.

4.3 Effect of Combustion Dynamics on Emissions and Operability (real time data):

Figure 10 shows the instantaneous NOx and CO emissions data collected with Configuration 1 (no area restriction). The sample is transferred from the probe to the sampling system in roughly 30 seconds. Though the low sampling frequency does not allow the sampling system to capture the frequency of oscillations, the flame oscillations are manifested in terms of oscillations in CO and NOx emissions as shown in Figure 11. There are roughly six zones of interest demonstrated on the figure.

Zone one corresponds to the quiet mode. It is apparent that no fluctuations in NOx and CO are present under this stable combustion mode. NOx climbs slightly as the flame temperature is increased. Zone 2 is equivalent to the intermittent mode which initiates at flame temperature near 2900 F. The impact on NOx and CO is enormous as shown in Figure 10. Zone 3 corresponds to a flame temperature of around 3000 F. This zone shows a tremendous increase in NOx while CO, on average, decreases slightly. Co and NOx fluctuations in this zone intensify in correspondence to the increase in the intermittent mode amplitude of oscillation. In Zone 4, NOx fluctuations are suppressed until a sudden drop in NOx and CO occurs at a flame temperature of 3250F. The CO equilibrates and the flame is steady. In an attempt to reduce the combustor head end flame temperature the fuel flow rate was decreased to 3200 F (Zone 6). This reduction is instantly associated with initiation of the intermittent oscillation mode and consequently with increased levels of NOx and CO as shown in Figure 10.

The results discussed in this section show that the fluctuations in NOx and CO emissions are good indicators of the amplitude of combustion instabilities. Also, the results show that approaching the conditions that trigger instabilities from higher or lower flame temperature has the same effect on the system behavior. The critical conditions that define the thermo-acoustic instabilities of the system are a property of the system and not dependent on the combustor loading sequence.

4.4 Effect of Radiation Heat Losses on Emissions:

As mentioned previously, the combustor liner and burner under investigation are identical to the quartz liner combustor investigated by Leonard and Correa [1] and Elkady et. al [6]. Figure 11 shows a comparison of the NOx emissions obtained from the current ceramic liner combustor (data collected at 179 Psia and 720F) to the data presented by Leonard and Correa quartz combustor (data collected at 152 Psia and 650F). The results are almost identical with slight discrepancy at high flame temperature due to a difference in residence time. Figure 13 shows a comparison of the data collected with the ceramic combustor at 258 Psia and 786 F with more recent data published by Elkady et. al (quartz liner with emissions data collected at 235 Psia and 715 F). Again, excellent agreement is noticed between both experiments.

Based on the agreement of quartz and ceramic liners, it can be concluded that the radiation heat losses from the flame has at most a small effect on the NOx emissions. It appears that at these flame temperatures, radicals are being produced and NOx production does not get quenched or slowed by radiation heat losses as long as the flame temperature is high enough to avoid reaction quenching. Also, the results show that the low amplitude sustained oscillations has no effect on emissions since there is excellent agreement between bot data sets.

5 CONCLUSIONS

Intermittent mode of oscillations significantly increases NOx and CO emissions. It is also function of pressure and it intensifies as the combustor operating pressure is increased. In cases where combustion oscillations affect emissions, the amplitude of emissions fluctuations is directly proportional to the amplitude of oscillations. Also, the conditions at which the oscillations are generated are shown to be independent of the combustor loading sequence, but rather to the system properties that get triggered at certain operating conditions.

This study shows that a 44% restriction in the exit area, changed dynamics boundary condition and was sufficient in this configuration to suppress the intermittent mode and restrict the acoustic communication between the combustor and the surrounding geometry. Low amplitude constant oscillations are shown to have little/ no effect on the combustor emissions.

Combustion emissions from ceramic and quartz liners are shown to almost identical. This points to limited/no effect of radiation heat losses on combustion emissions

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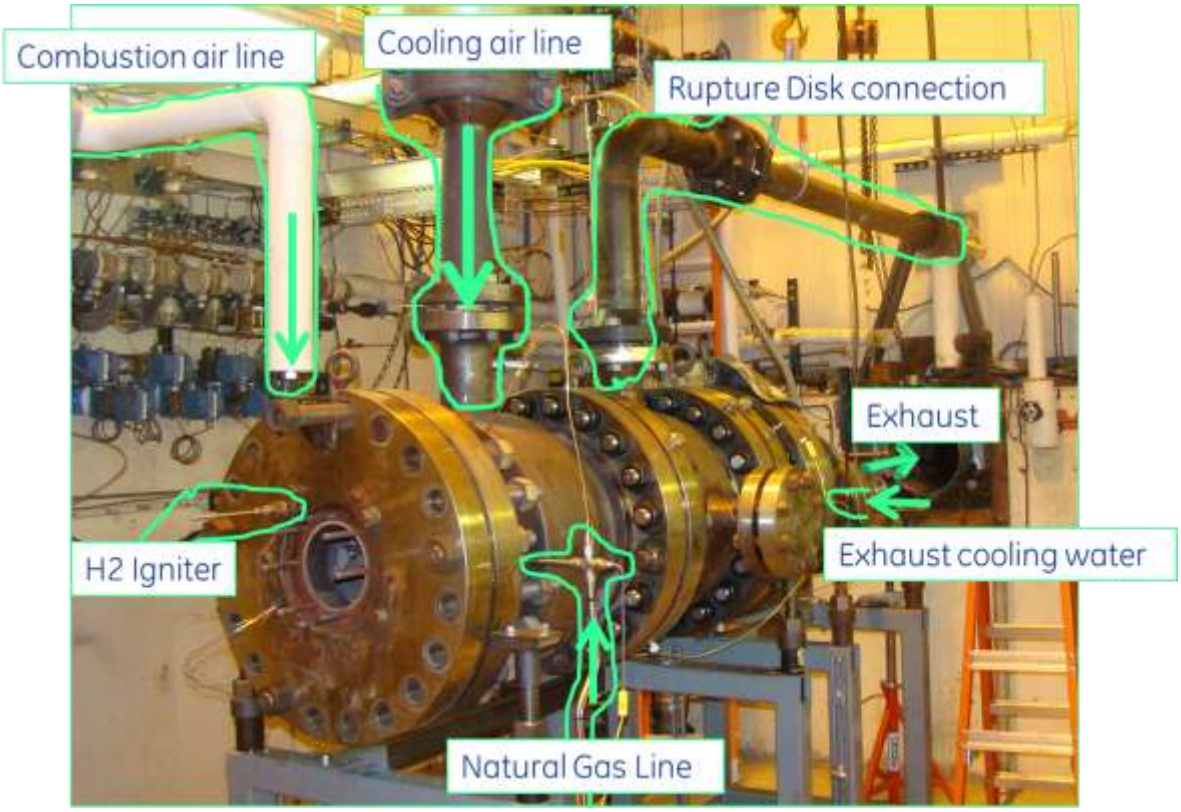

Figure 1. Pressure vessel and air/fuel connections.

Figure 2. Air /Fuel mixing tube with surface temperature monitoring.

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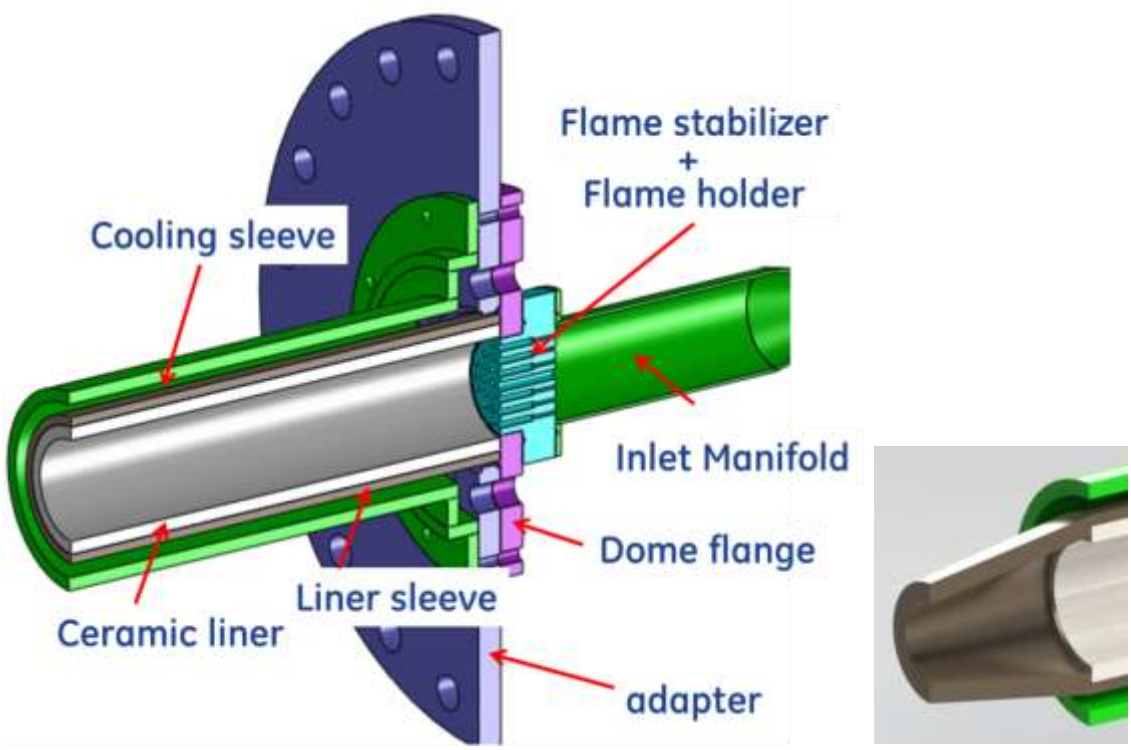

Figure 3. Ceramic Combustor Assembly. Exit area restriction is shown to the right.

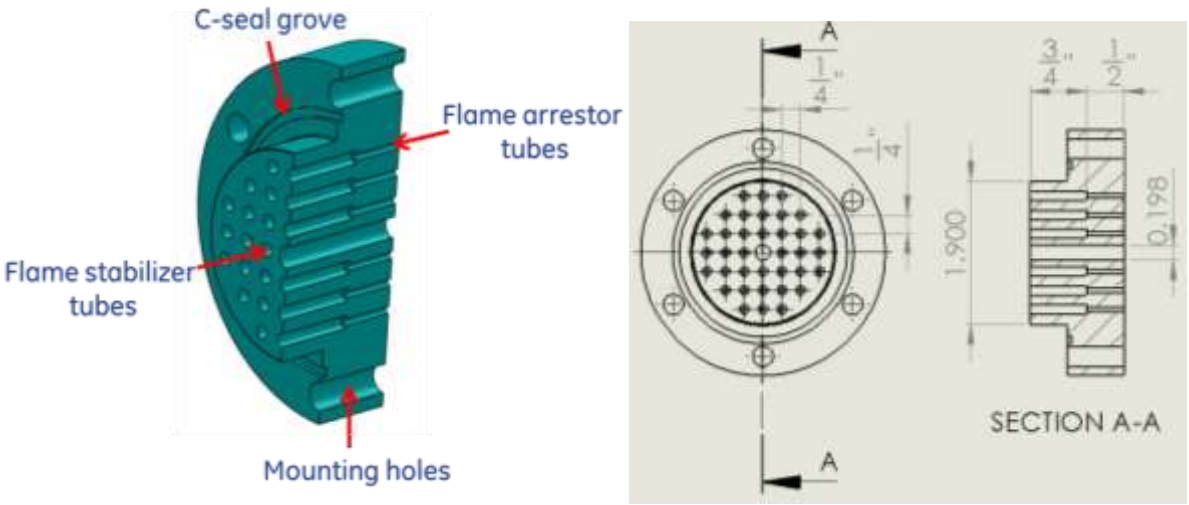

Figure 4. Perforated flame burner.

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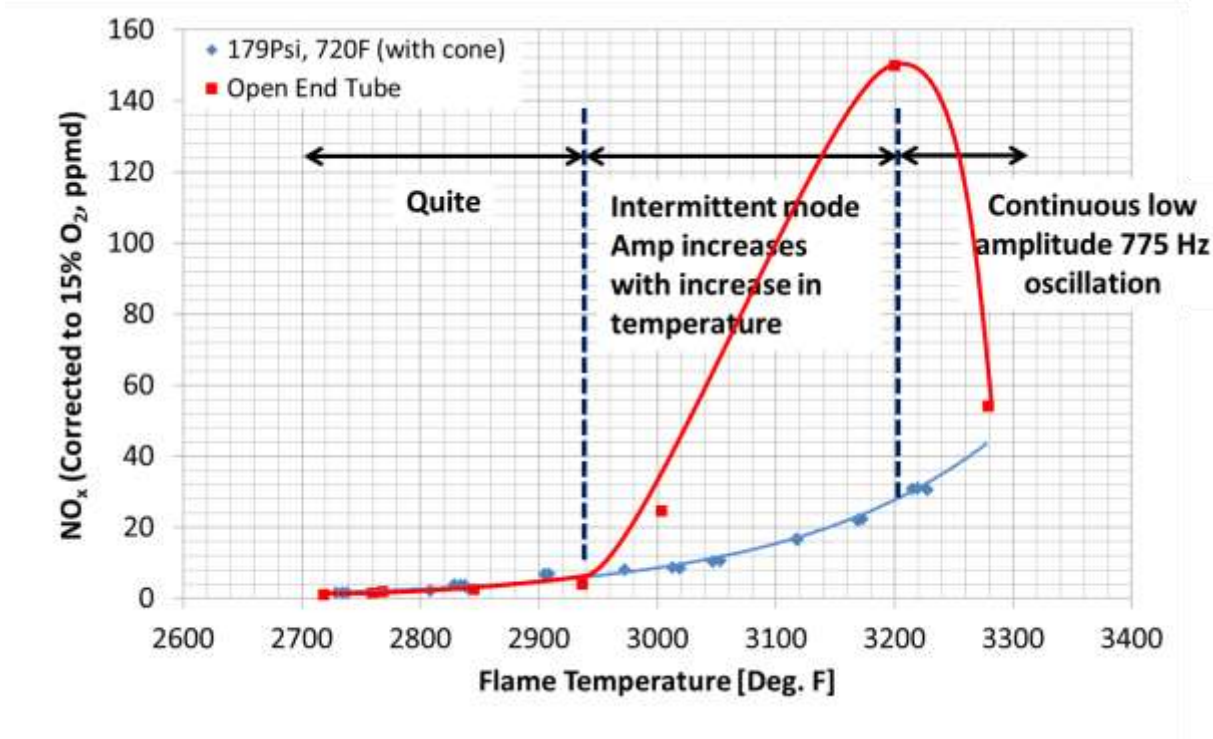

Figure 5. NOx15 emissions vs. flame temperature for both combustor configurations.

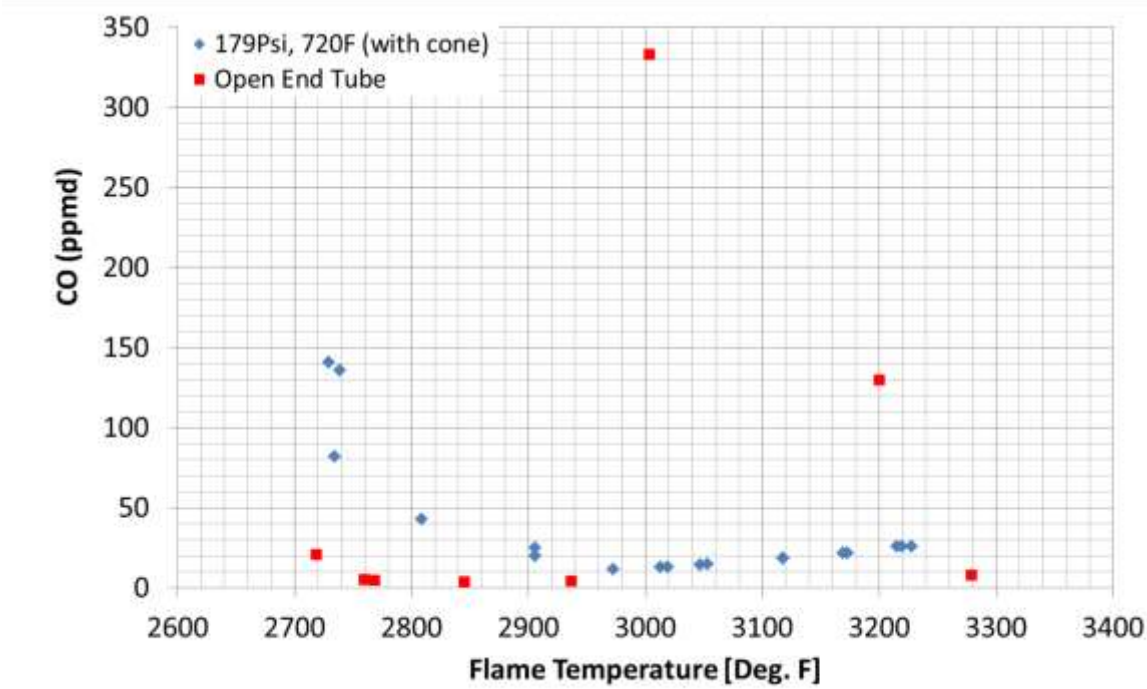

Figure 6. CO emissions vs. flame temperature for both combustor configurations.

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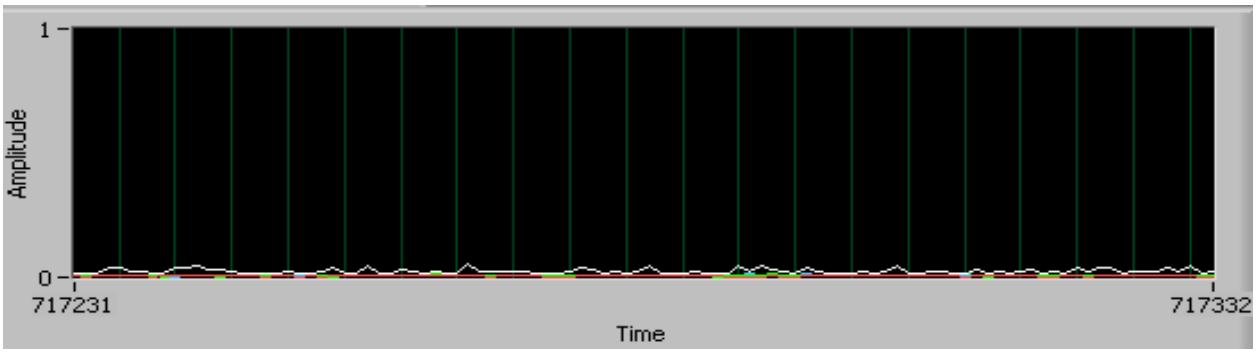

Figure 7. Instability amplitude vs. time (Quite mode).

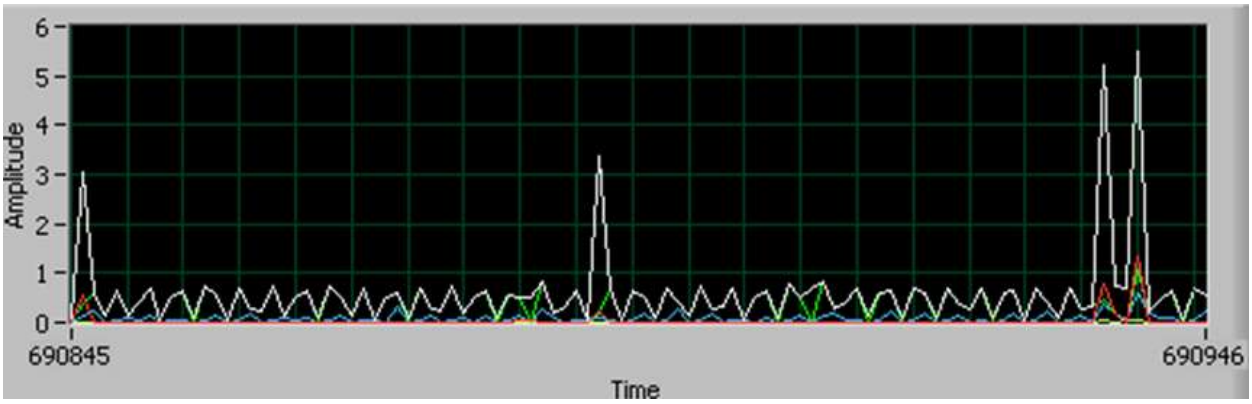

Figure 8. Instability amplitude vs. time (Intermittent mode).

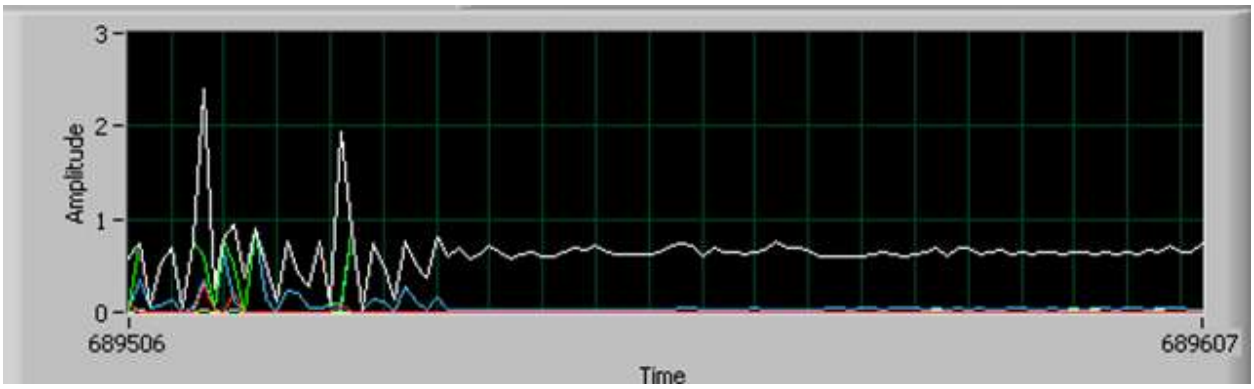

Figure 9. Instability amplitude vs. time (transition from intermittent to continuous low amplitude oscillation).

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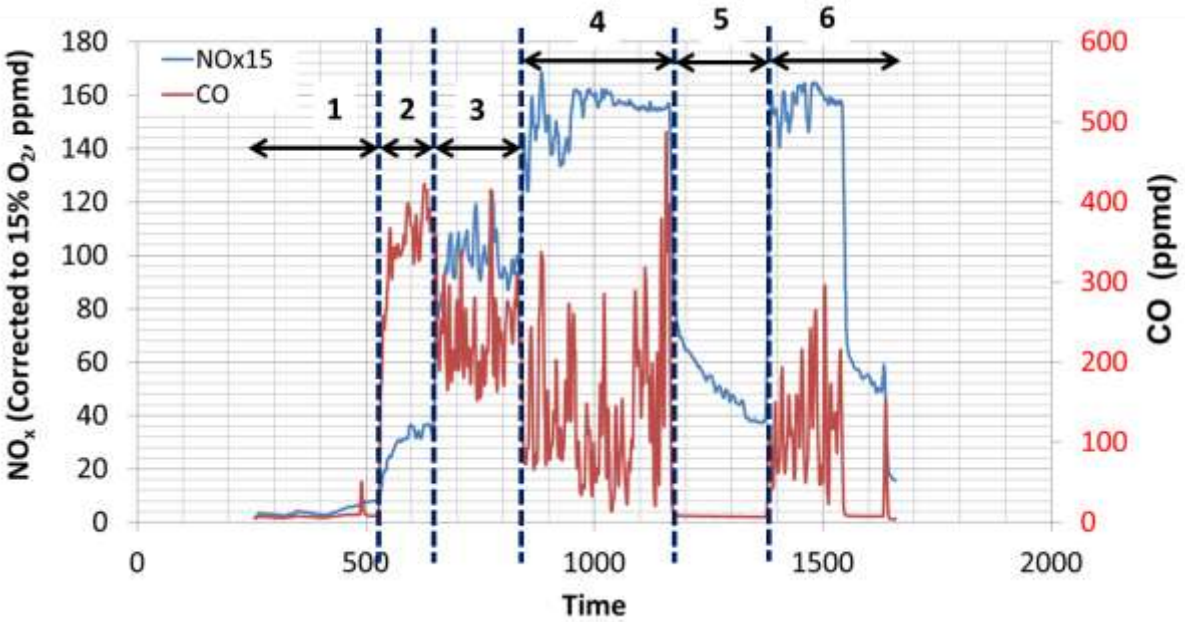

Figure 10. Instantaneous NOx and CO emissions as function of time (Data for Configuration 1).

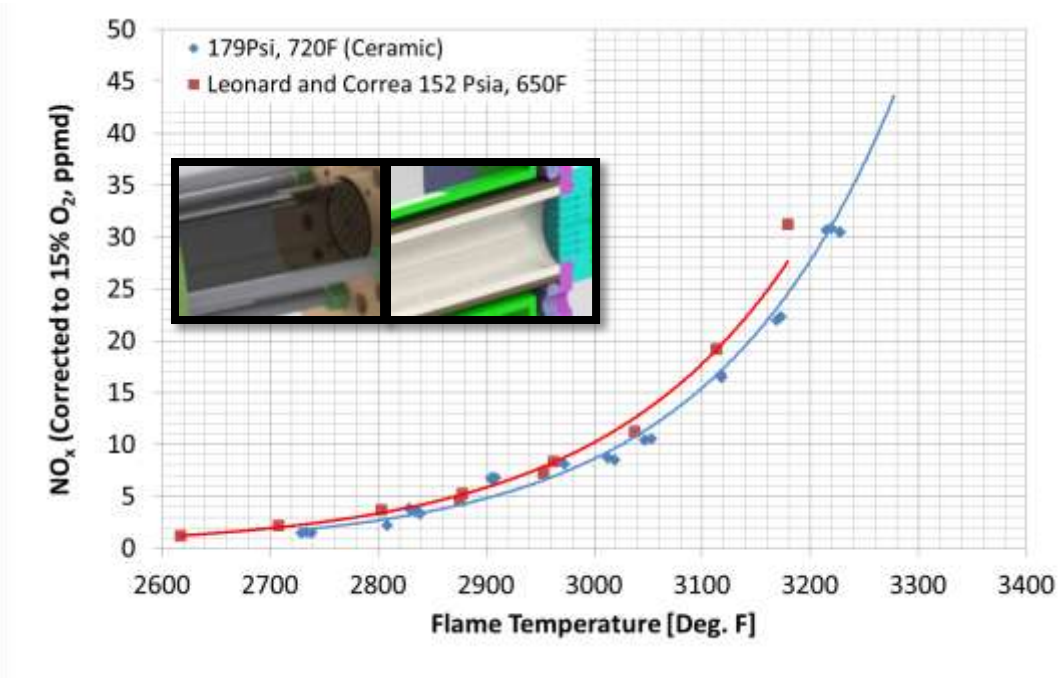

Figure 11. Comparison of NOx emissions from Configuration 2 with Leonard and Correa.

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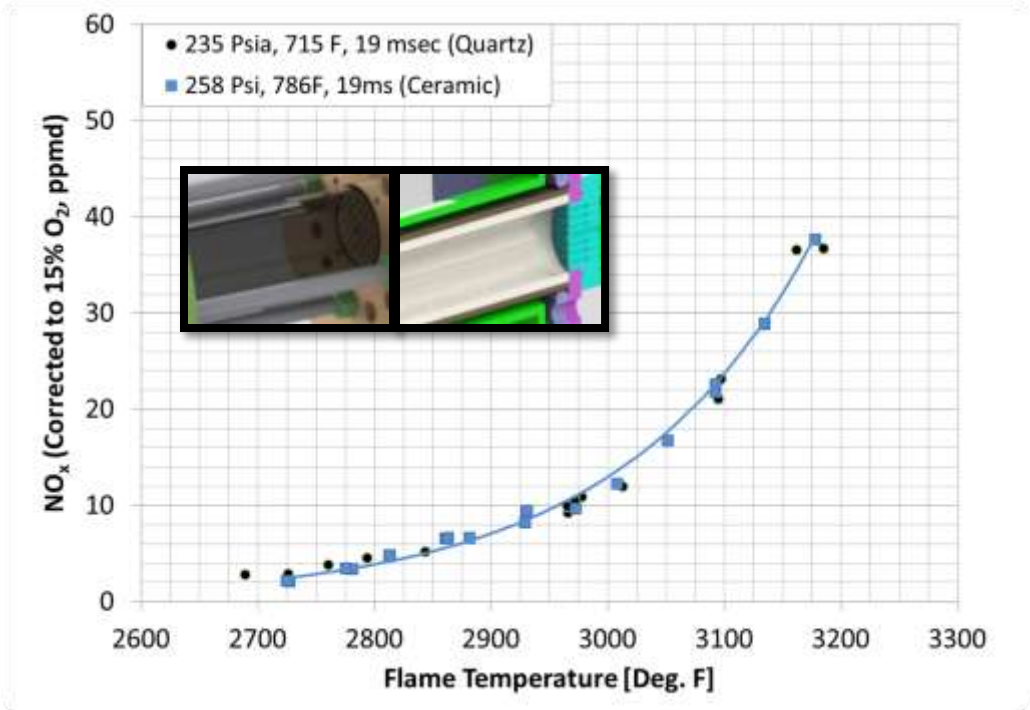

Figure 12. Comparison of NOx emissions from Configuration Two with Elkady et. al.

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